

Database Project – Step 2

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1)

Artists: ArtistID, Aname, Acity, Aemail, AbirthYear, Abio, Aphone, Awebsite, AsocialMedia, isActive, #Adiscipline (refers to Name in Discipline)

Artworks: ArtworkID, AWtitle, AWtype, AWprice, AWstatus, AWcreationYear, AWmedium, AWdimensions, AWdescription, #AWartist, #AWtag (refers to AWtname)

ArtworkTag: AWtname

Booking: BookingID, BookingDate, PaymentStatus, #Member (refers to MemberID), #Workshop (refers to Title in Workshops)

Exhibitions: ExhibitionID, EXtitle, EXstartDate, EXtheme, EXendDate, EXdescription, EXcuratorName, #EXartworks (refers to ArtworkID), #Gallery (refers to GalleryID)

Workshops: WorkshopID, WStitle, WSdate, WSprice, WSlevel, DurationMinutes, MaxParticipants, WSlocation, WSdescription, #WSinstructor (refers to ArtistID)

CommunityMember: MemberID, CMname, CMemail, CMcity, CMbirthYear, CMphone, MembershipType, #CMbookings (refers to BookingID), #CMreviews (refers to ReviewID), #FavoriteDiscipline (refers to Name in Discipline)

Gallery: GalleryID, Gname, Gaddress, Grating, GownerName, GopeningHours, GcontactPhone, Gwebsite, #Gexhibition (refers to ExhibitionID)

Discipline: Dname

Review: ReviewID, Rating, Comment, ReviewDate, #Reviewer (refers to MemberID), #Artwork (refers to ArtworkID)

Functional dependencies:

ArtistID -> Aname, Acity, Aemail, AbirthYear, Abio, Aphone, Awebsite, AsocialMedia, isActive, Dname

ArtworkID -> AWTitle, AWType, AWPrice, AWStatus, AWCreationYear, AWMedium, AWDimensions, AWDescription, ArtistID, AWName (ArtworkTag)

BookingID -> BookingDate, PaymentStatus, Member (CommunityMember), Artwork (ArtworkID)

ExhibitionID -> ExTitle, ExStartDate, ExTheme, ExEndDate, ExDescription, ExCuratorName, ArtworkID (Artworks)

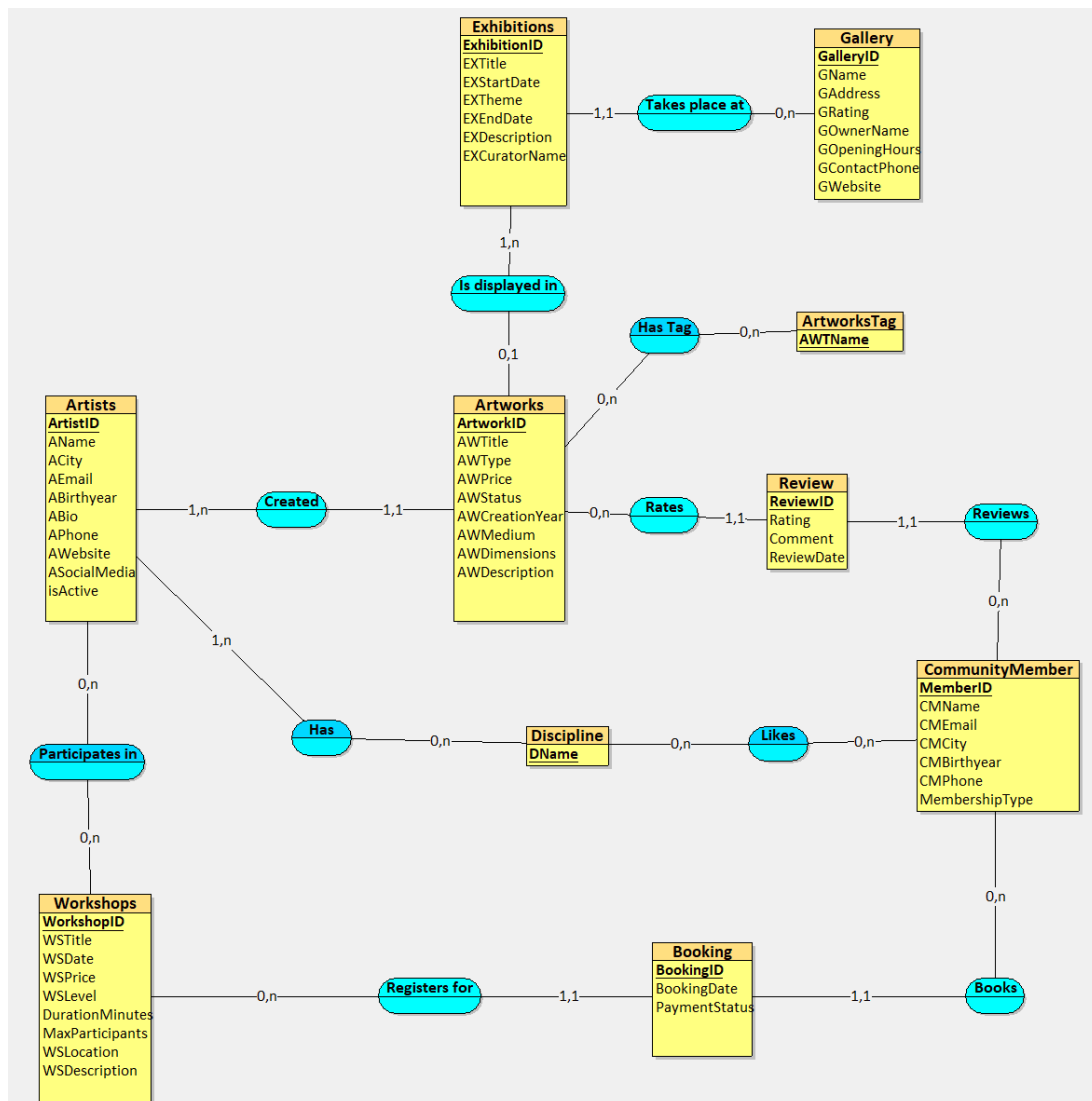
WorkshopID -> WSTitle, WSDate, WSPrice, WSLevel, DurationMinutes, MaxParticipants, WSLocation, WSDescription

MemberID -> CMname, CMemail, CMcity, CMbirthYear, CMphone, MembershipType, #CMbookings (refers to BookingID), #CMreviews (refers to ReviewID), #FavoriteDiscipline (refers to Name in Discipline)

GalleryID -> GalleryID, Gname, Gaddress, Grating, GownerName, GopeningHours, GcontactPhone, Gwebsite, #Gexhibition (refers to ExhibitionID)

ReviewID -> Rating, Comment, ReviewDate, Reviewer (MemberID), Artwork (ArtworkID)

2)



3)

Artists = (ArtistID INT, Aname VARCHAR(50), Acity VARCHAR(50), Aemail VARCHAR(50), AbirthYear DATE, Abio VARCHAR(200), Aphone VARCHAR(20), Awebsite VARCHAR(100), AsocialMedia VARCHAR(50), isActive BOOLEAN);

ArtworksTag = (AWTName VARCHAR(25));

Workshops = (WorkshopID INT, WStitle VARCHAR(100), WSdate DATE, WSprice DECIMAL(3,2), WSlevel VARCHAR(12), DurationMinutes INT, MaxParticipants INT, WSlocation VARCHAR(100), WSdescription VARCHAR(200));

CommunityMember = (MemberID INT, CMname VARCHAR(50), CMemail VARCHAR(50), CMcity VARCHAR(50), CMbirthYear DATE, CMphone VARCHAR(20), MembershipType VARCHAR(7));

Gallery = (GalleryID INT, Gname VARCHAR(50), Gaddress VARCHAR(50), Grating DECIMAL(1,1), GownerName VARCHAR(50), GopeningHours VARCHAR(100), GcontactPhone VARCHAR(20), Gwebsite VARCHAR(100));

Discipline = (DName VARCHAR(50));

Booking = (BookingID INT, BookingDate DATE, PaymentStatus VARCHAR(9), #WorkshopID, #MemberID);

Exhibitions = (ExhibitionID INT, EXtitle VARCHAR(100), EXstartDate DATE, EXtheme VARCHAR(50), EXendDate DATE, EXdescription VARCHAR(200), EXcuratorName VARCHAR(50), #GalleryID);

Artworks = (ArtworkID INT, AWtitle VARCHAR(100), AWtype VARCHAR(25), AWprice DECIMAL(9,2), AWstatus VARCHAR(9), AWcreationYear INT, AWmedium VARCHAR(20), AWdimensions VARCHAR(20), AWdescription VARCHAR(200), #ExhibitionID, #ArtistID);

Review = (ReviewID INT, Rating DECIMAL(1,1), Comment VARCHAR(500), ReviewDate DATE, #ArtworkID, #MemberID);

Has = (#ArtistID, #DName);

Likes = (#MemberID, #DName);

Has_Tag = (#ArtworkID, #AWTName);

Participates_in = (#ArtistID, #WorkshopID);

Right off the bat, we already are in 1NF since all of the attributes are atomic. In addition, every single attribute depends on their respective primary keys so this is also in 2NF and this can also be checked thanks to the functional dependencies we have written somewhere above. Finally, also thanks to those dependencies we know that all of the

attributes don't depend on non-key attributes so this fits 3NF. Because we have carefully crafted these tables we didn't have to change them to be in 3NF.